Aadit Kannan

Selected Hardware Projects

Actuators, HV accumulator hardware, PCB test instrumentation, GD&T/manufacturing, and robot mechanisms. Full portfolio at aaditkannan.com/projects.

CONTACT

WEBSITE
aaditkannan.com

PROJECTS
aaditkannan.com/projects

EMAIL
aaditkannan@berkeley.edu

GITHUB
github.com/aaditkannan

LINKEDIN
linkedin.com/in/aaditkannan

PORTFOLIO CONTENTS

02 Wolfrom Architecture

03 Wolfrom Prototype + Metal Revision

04 Formula Electric 588V Accumulator

05 Accumulator Materials + Validation

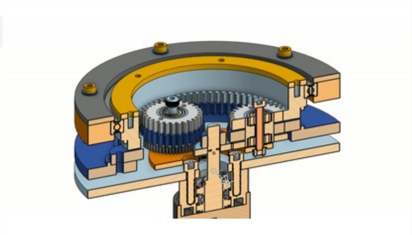
06 Nanosecond Spin Transport Pulse Generator Board

07 Pulse Generator V1 Learning

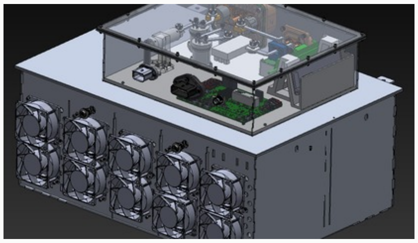
08 Custom Toolbox Design + Manufacturing

09 FIRST Robotics Hardware

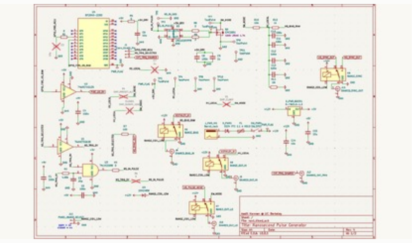
10 Tools + Methods



Actuator section render



Accumulator attic CAD

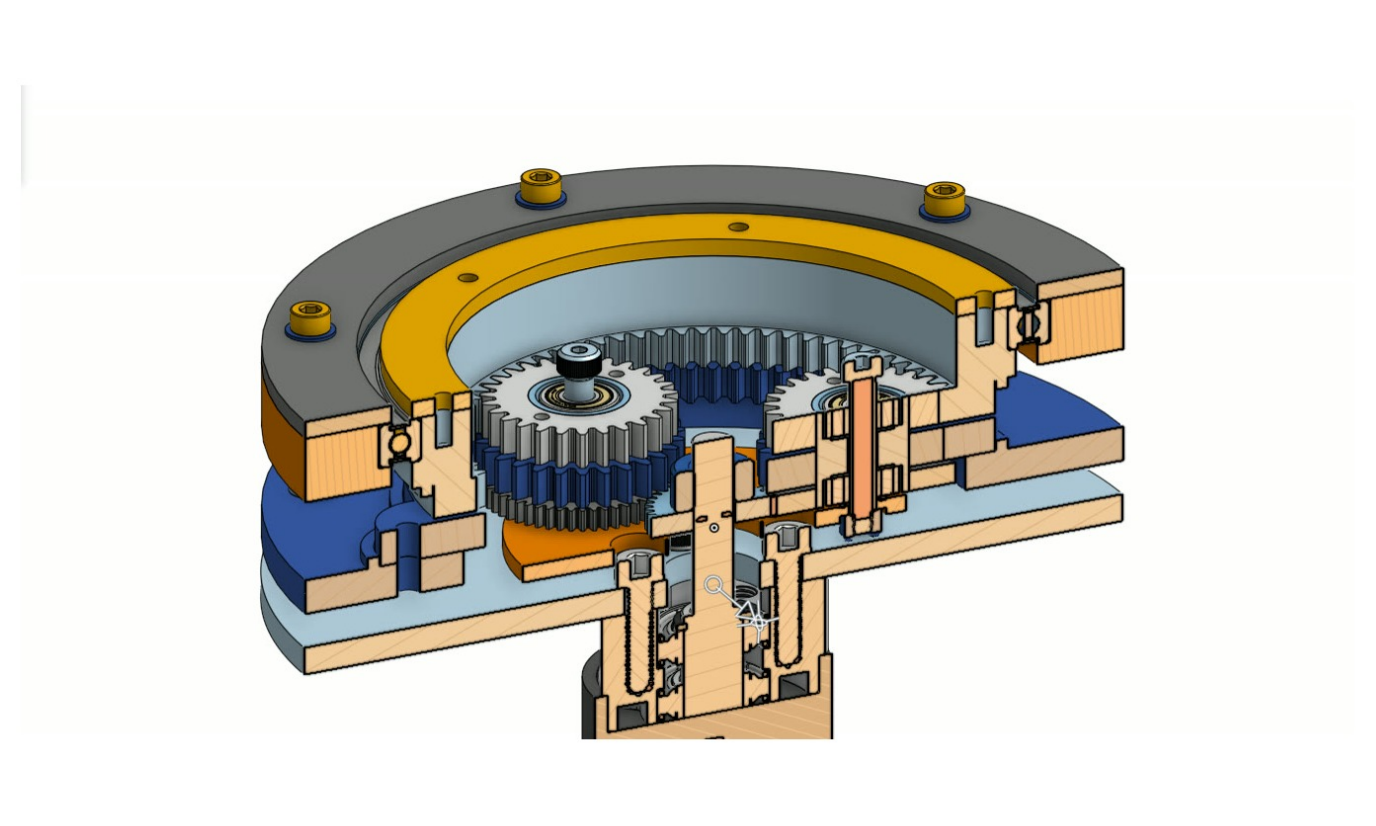


Pulse-generator schematic

Wolfrom Compound Planetary Actuator

May 2026 - Present / Robotics Hardware

3-stage 50:1 compound Wolfrom actuator for backdrivable robotic joints, built around compact packaging, reduced circulating power losses, split-gear preloading, and prototype-to-metal validation.



CAD section showing the compact axial stack: pre-reduction stage, compound planets, fixed ring, output ring, bearing support, output hub, and motor interface.

DESIGNED

- Full actuator stackup: sun, compound planets, fixed ring, output ring, carrier, bearings, shafts, and output hub.
- 3-stage compound layout in one compact axial package instead of stacking multiple conventional planetary gearboxes.
- Split output ring, bolt access, dowel alignment, bearing support, preload strategy, and assembly order.
- Built FDM prototypes to validate meshing, stack height, bearing placement, and assembly before moving into metal hardware.

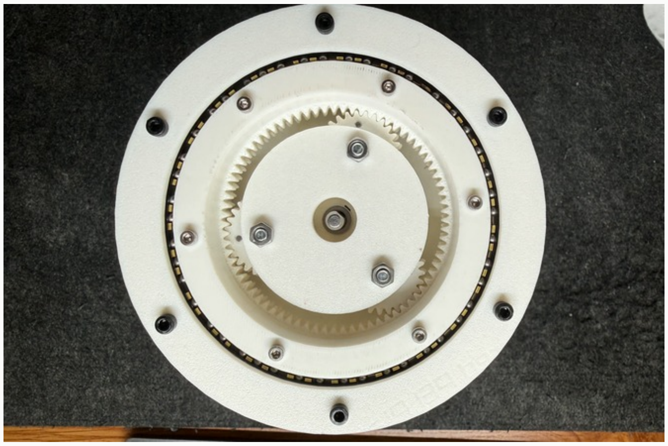
GEARBOX NOTES

- Current tooth counts give 555/11, or about 50.45:1 reduction.
- Maximized pre-reduction to reduce circulating power losses in the Wolfrom differential stage.
- Targeting 80% efficiency and near-zero backlash through friction reduction, bearing/mesh choices, and split-gear preloading.
- Primary loss paths are ring-mesh friction, planet bearing drag, load sharing, and rough printed tooth surfaces.

Wolfrom Compound Planetary Actuator

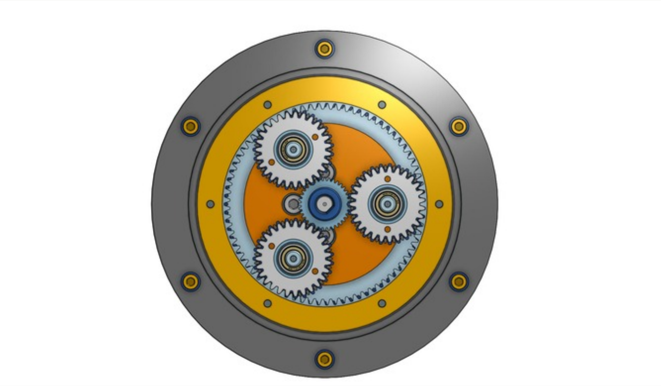
May 2026 - Present / Robotics Hardware / Prototype + Metal Revision

FDM prototypes are being used to validate meshing, stack height, bearing support, output-ring splitting, and assembly sequence while the CNC aluminum / wire-EDM revision and dyno setup are developed.

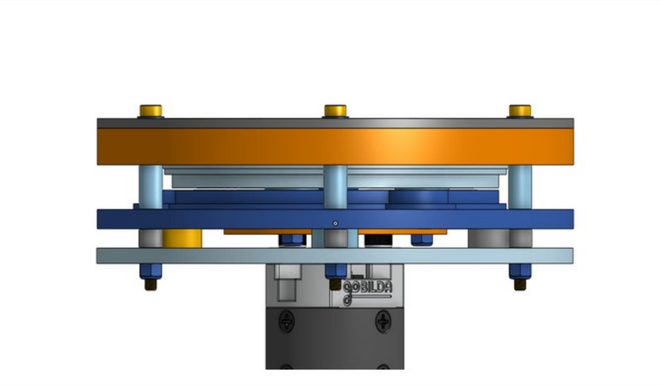


Printed prototype used for gear mesh, bearing placement, and stack-height checks.

Bench assembly and fit-checking during gear timing and stackup validation.



Top view: three compound planets, output ring, fastener pattern, and center sun.



Side view: plate stack, motor interface, standoffs, and axial package.

CURRENT PROTOTYPE

- Built and fit-checked FDM actuator prototypes for gear meshing, assembly access, and stack alignment.
- Working on split-gear preload to reduce backlash; no final backlash number yet.
- Printed gears are being used for geometry and assembly validation, not final load-rated actuator performance.

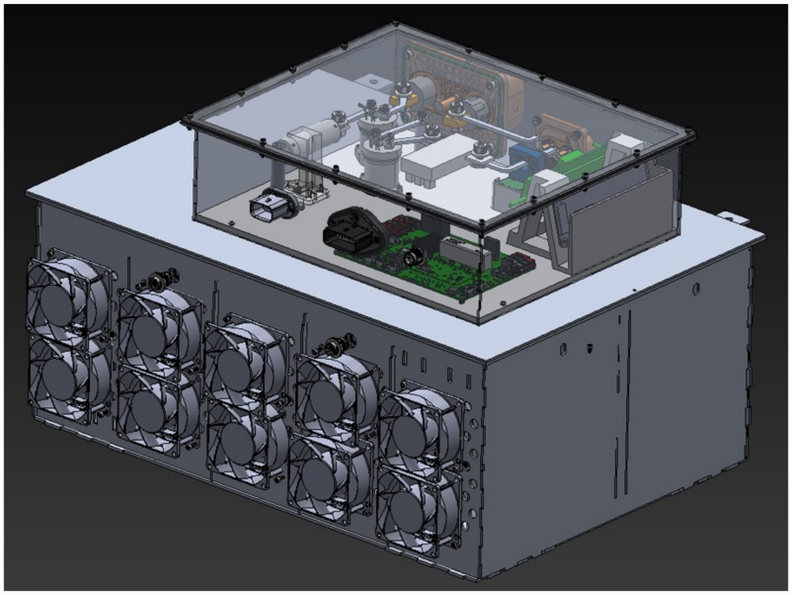
METAL REVISION + TEST

- Simulating gear macro-geometry in KISSsoft, including profile shifts and tip relief.
- Manufacturing CNC aluminum structure with wire-EDM ring gears for the next hardware revision.
- Building dyno setup for torque, efficiency, breakaway torque, backdrive feel, and load-behavior testing.

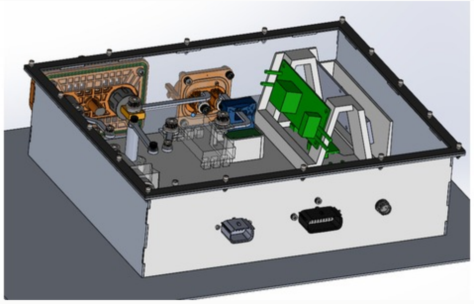
Formula Electric 588V Accumulator

Sep 2025 - Present / High-Voltage Packaging

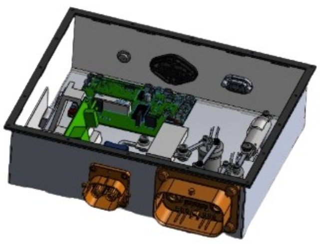
Accumulator attic and enclosure work for Berkeley Formula Electric, integrating HV electronics, busbars, insulation, cooling, water sealing, and service access inside a constrained pack volume.



588V accumulator CAD showing the HV electronics attic, fan wall, service access, and enclosure packaging.



Electronics attic layout: board, connector, busbar, and mechanical interface packaging.



Internal attic CAD view with board/connector access and enclosure clearances.

BUILT

- Packaged the HV attic: boards, busbars, fans, fasteners, service interfaces, and inspection access.
- Cut attic mass by 18% while keeping clearances, cooling paths, and fastener access workable.
- Designed around a 588V pack, water sealing, dielectric isolation, and SES/ESF review needs.

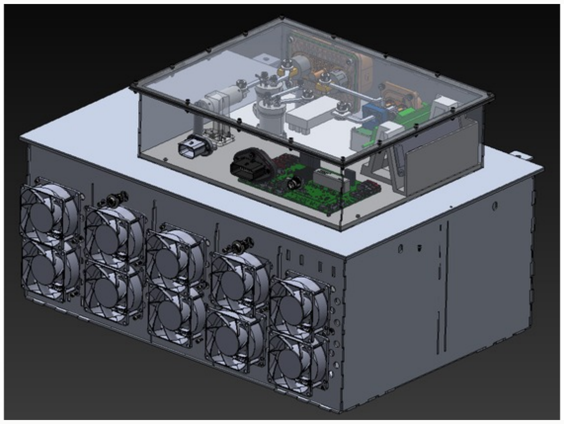
CONSTRAINTS

- Busbar material and package design for 80A peak discharge without making the attic impossible to service.
- FSAE load cases, HV isolation, rain-test sealing, airflow, and board access all compete for the same space.
- Transparent/inspectable surfaces matter because serviceability is part of safety.

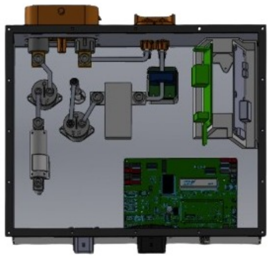
Formula Electric 588V Accumulator

Sep 2025 - Present / High-Voltage Packaging / Materials + Validation

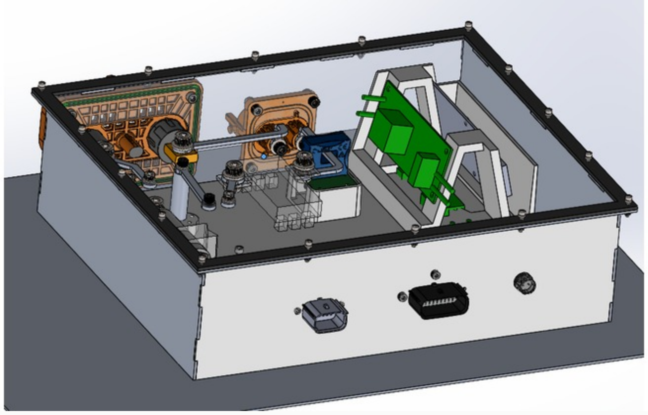
Materials, manufacturing methods, and validation checks for turning the accumulator CAD into hardware that can meet competition inspection and reliability requirements.



Overall accumulator CAD context: attic, enclosure, service access, and internal packaging.



Top-down attic CAD view used to check electronics layout and access.



Detailed accumulator subassembly view for board, connector, and internal support packaging.

MANUFACTURING PLAN

- Laser-cut aluminum panels, TIG welded structure, waterjet polycarbonate floor, and waterjet neoprene gasket.
- Polycarbonate floor gives electrical isolation and visual inspection without disassembling the pack.
- Nomex 410 for dielectric strength, temperature margin, and flame resistance near HV components.

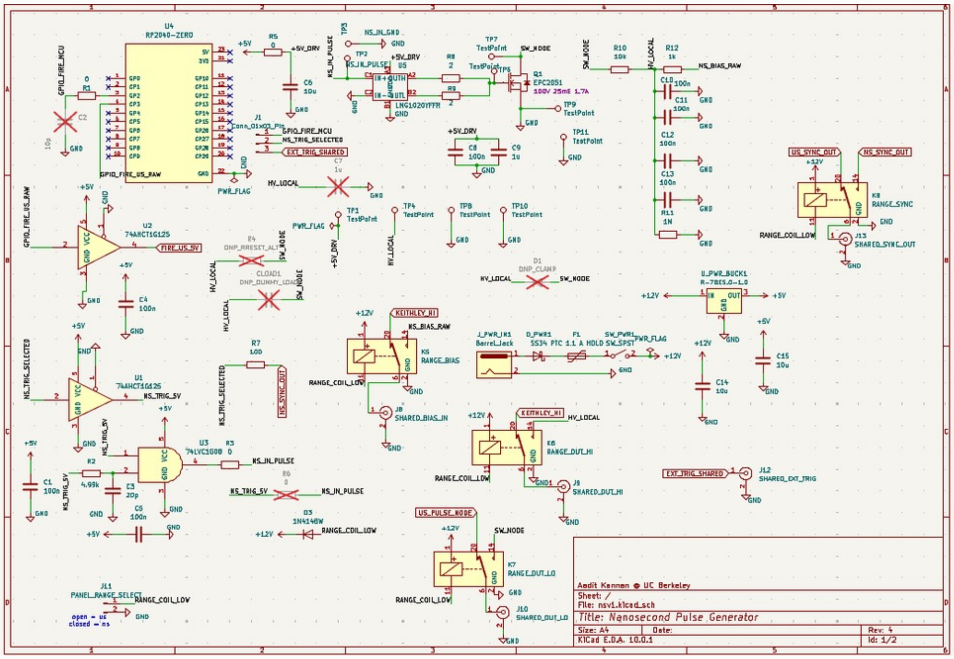
CHECKS

- FEA against 40g lateral, 40g longitudinal, and 20g vertical load cases.
- Checked gasket compression, water ingress, dielectric margin, service access, fastener access, and fan airflow together.
- Target gates include 1500V AC dielectric testing and IP65-style water ingress testing before event use.

Nanosecond Spin Transport Pulse Generator Board

May 2026 - Present / Research Hardware / PCB

Board for fast electrical testing of $BiFeO_3$ devices in Ramesh Lab. The current design is moving from a first-generation microsecond pulser toward cleaner ns/us triggering, grounded coax/BNC I/O, and repeatable scope/SMU measurements.



Current nanosecond pulse-generator schematic. Shown as the main artifact because it is the current board revision.

BENCH INTERFACE

Python/RP2040 trigger -> driver and switching stage -> coax/BNC output -> device or probe station -> oscilloscope and source-measure capture.

DESIGNED

- Dual-range pulse architecture around current BFO testing near 30V, with future lower-voltage sweeps planned.
- GaN switching path, trigger/sync routing, grounded lab I/O, and source-measure bias interface.
- Layout organized around probe-station use, dummy-load bring-up, and scope visibility.

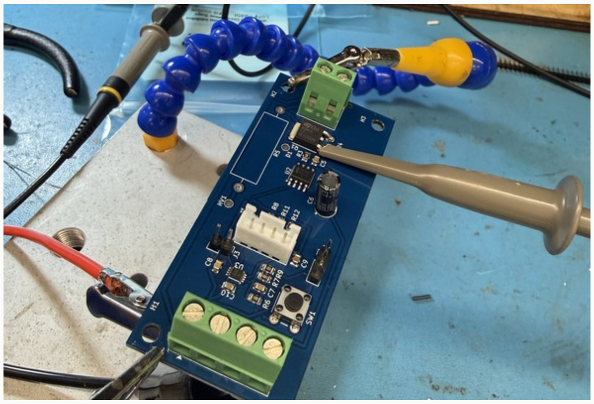
DEVICE-SIDE NOTES

- What matters is the voltage that actually lands across the BFO device, not just the connector setting.
- Measured current can mix polarization switching, leakage, dielectric capacitance, and fixture parasitics.
- No final rise-time, jitter, device result, or amplitude-stability claim yet; bring-up is still in progress.

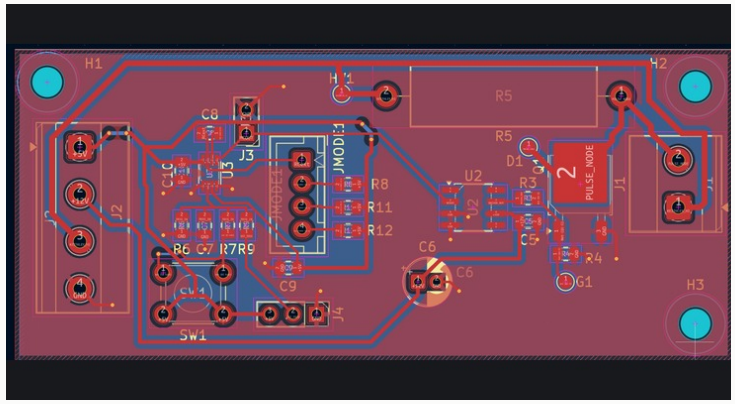
Nanosecond Spin Transport Pulse Generator Board

May 2026 - Present / Research Hardware / Board + V1 Learning

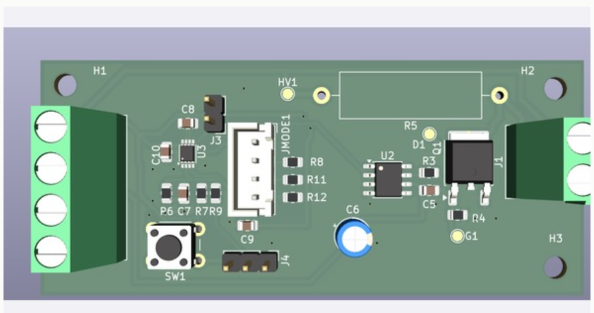
V1 microsecond hardware established the initial bench workflow. The current revision carries that learning into cleaner triggering, lab I/O, grounding, and measurement flow.



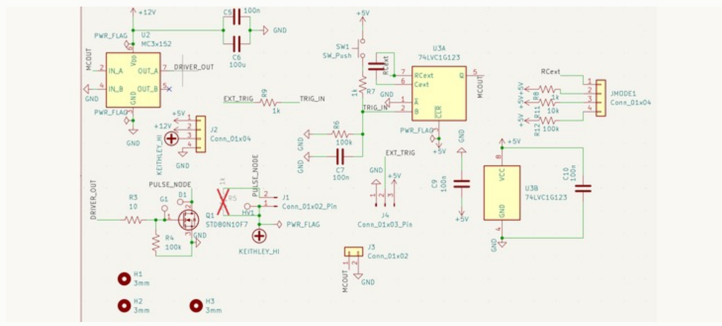
V1 microsecond board during bench probing. Useful as a learning artifact, not the current ns revision.



V1 PCB layout used to inspect trace routing, connector placement, and returns.



V1 rendered PCB for physical integration reference.



V1 microsecond schematic, included only as a predecessor to the current ns/us board.

V1 EXPOSED

- Clip-lead power and weak lab I/O made the first board awkward around real bench equipment.
- Triggering needed to be tied into Python/RP2040 control, not treated as an isolated pulse node.
- LTspice transient work informed switching timing and V1 bring-up, while the current revision is documented through schematic/layout and bench validation artifacts.
- Grounding and connector strategy had to be designed into the instrument rather than left to bench improvisation.

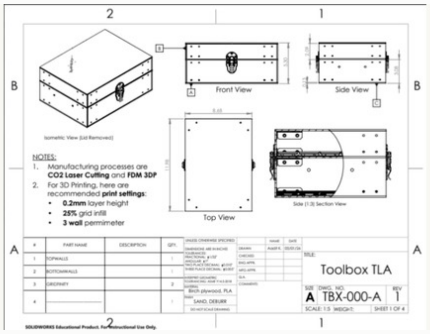
CURRENT REVISION ADDS

- Nanosecond and microsecond paths in one cleaner board.
- Coax/BNC-style outputs, shared sync/trigger handling, source-measure biasing, and clearer return paths.
- Bring-up order: rails, trigger logic, dummy loads, switching behavior, then device-side fixture measurements.

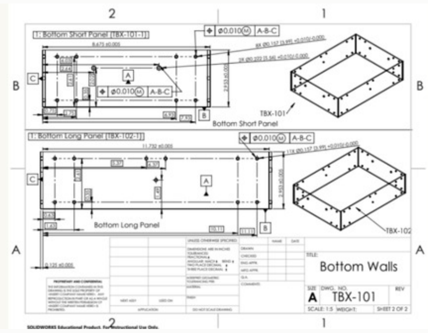
Custom Toolbox Design + Manufacturing

Jan 2026 - May 2026 / Mechanical Design

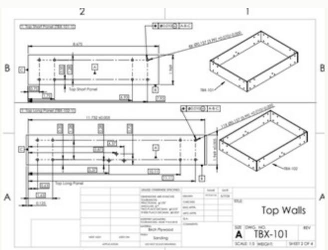
Mechanical design project centered on drawings, datums, tolerances, fabrication choices, and fit-up inspection for a laser-cut/FDM toolbox assembly.



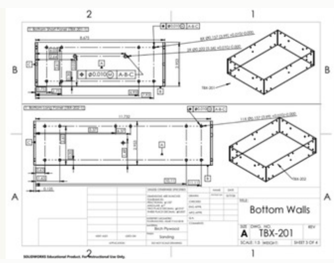
Assembly drawing defining lid, latch, hinge, body, and insert interfaces.



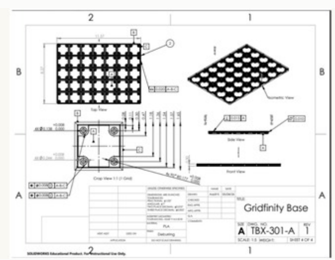
Bottom wall drawing with datums, critical dimensions, and positional tolerance callouts.



Top wall drawing for lid/body alignment.

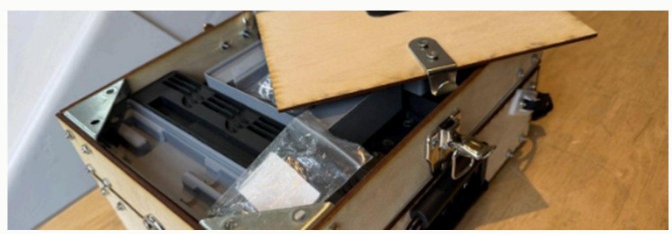


Revised bottom wall drawing with updated hole pattern control.



Gridfinity insert drawing with pocket and fastener tolerances.

Closed prototype fit-up with hinges, latches, handle, and fasteners.



Open prototype showing interior fit-up, inserts, hinge clearance, and lid alignment.

DESIGNED

- Production-style drawings using ASME Y14.5-2018 GD&T conventions.
- Datum scheme, position tolerances, fastener clearances, hinge/latch interfaces, and lid closure.
- Laser-cut plywood body with FDM/Gridfinity-style inserts and visible fastened joints.

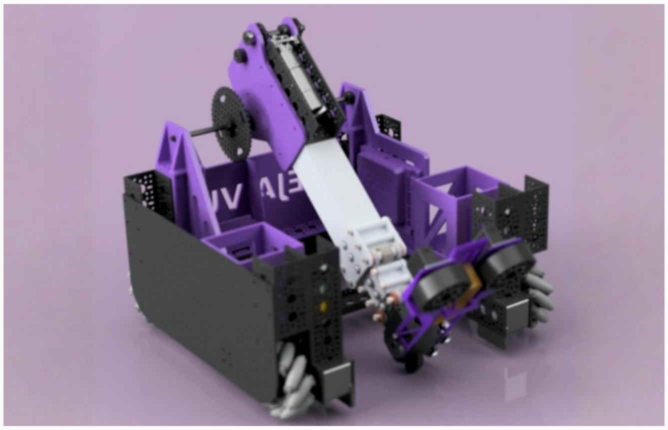
INSPECTED

- Fit-up checked with calipers, fasteners/gauge pins, squareness checks, lid closure, and hinge/latch alignment.
- Prototype used laser-cut plywood and FDM inserts; the engineering focus was drawing release, datum strategy, tolerancing, fabrication, and fit-up inspection.
- Prototype photos show the fabricated assembly used for hinge, latch, handle, fastener, and interior fit-up checks.

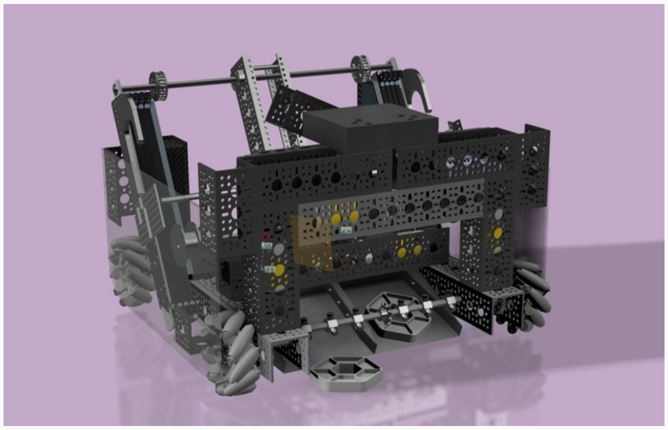
FIRST Robotics Hardware

Aug 2021 - Jun 2025 / Robotics Hardware / Controls

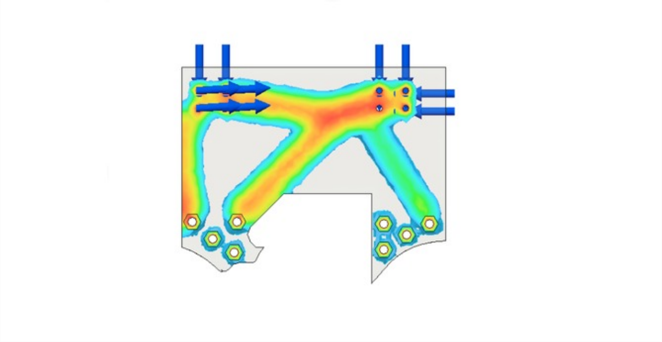
Four seasons of FTC hardware: 8 robots, 500+ part CAD assemblies, mechanism iteration under match constraints, Java controls, and three Robot Design Awards.



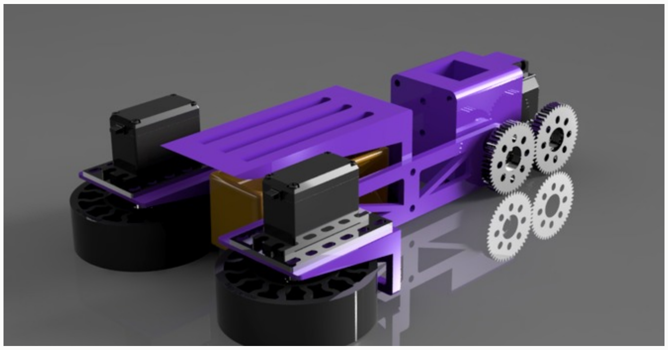
INTO THE DEEP robot CAD with drivetrain, active intake, transfer path, lift, and end effector.



CENTERSTAGE robot CAD assembly with pixel scoring mechanisms.



FTC bracket FEA used to check stress paths around arm/hang loading and fastener interfaces.



High-iteration intake subsystem CAD from Deja Vu INTO THE DEEP.

BUILT ACROSS SEASONS

- Mecanum drivetrains, active intakes, transfer paths, arms/lifts, claws, end effectors, hang mechanisms, and scoring systems.
- FEA on arm pivots, hang brackets, and custom aluminum/printed parts to check stress, safety factor, deflection, and fastener load paths.
- Parts made through printing, laser-cut plates, CNC aluminum, and normal shop fabrication.

CONTROLS-AWARE DESIGN

- Java TeleOp with field-centric mecanum, encoder PID loops, mechanism sequencing, driver feedback, and autonomous routines.
- Hardware and driver code were tuned together so mechanisms remained usable under match timing, driver-control, and service constraints.
- Mentored 30+ members in CAD, Java, mechanism iteration, and manufacturing workflow.

Tools + Methods

Current / Engineering Toolkit

A compact map of the tools and methods that show up across the project pages.

CAD / MECHANICAL

- SolidWorks
- Onshape
- Fusion 360
- GD&T
- DFM/DFA
- FEA
- mechanism packaging
- tolerance stackups

MANUFACTURING

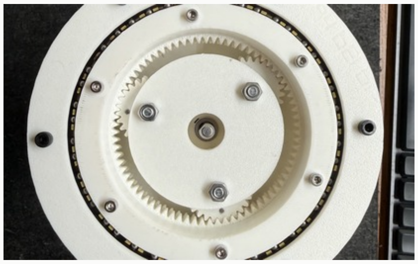
- mill
- lathe
- waterjet
- laser cutter
- TIG
- FDM/SLA
- Markforged
- fastener fit-up

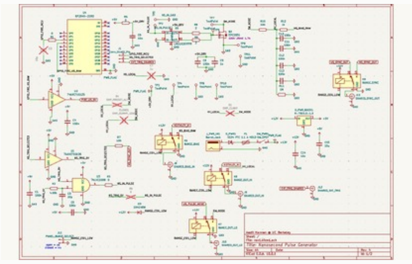
ELECTRONICS / TEST

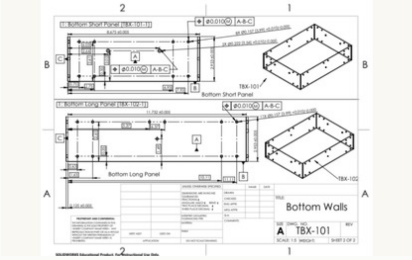
- KiCad
- LTspice
- oscilloscope
- source-measure
- RP2040/Python
- coax/BNC interfaces
- microcontrollers

SOFTWARE / DATA

- Python
- MATLAB/Simulink
- Java
- React/TypeScript
- Node.js
- PostgreSQL
- JupyterLab
- LabVIEW


Actuator prototype


PCB schematic



GD&T drawing